Zero- Versus Infinite-Temperature Damping in Variational Quantum Circuits

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Randomized compiling makes hardware noise easier to handle by twirling it into Pauli noise, but it also removes the pull of amplitude damping toward $|0\rangle$. This has been reported to make variational models worse, and reversing the direction of the damping to make them worse still. We ask what these effects really are. Comparing amplitude damping with its Pauli twirl, which is the same damping at infinite temperature, we find that the twirl mainly flattens the measured expectation values across inputs. A trained output scale undoes most of the loss in accuracy, so the models learn just as well, but the price returns as measurement shots. Under the twirl the flattened values must be measured far more often, while under amplitude damping a floor keeps them resolvable. This floor vanishes as the temperature rises, and at realistic damping strengths it matters only in very deep circuits. We also show that reversing the damping is usually a gauge freedom that changes only the parameters, and that the reported direction effect arises only because the damping sits inside a decomposed two-qubit gate.

I. INTRODUCTION

Variational quantum algorithms and quantum machine learning models are built from parameterized circuits and are meant to run on noisy near-term devices [1–3]. They can only be trained if their outputs change with the parameters. In deep random circuits, this change vanishes exponentially with the number of qubits, a problem known as a barren plateau [4–6], and noise makes it worse. Unital noise drives the state toward the maximally mixed state, so local expectation values decay exponentially with depth [7, 8]. Non-unital noise behaves differently. Amplitude damping (AD) drives the state toward a pure state and avoids barren plateaus for local observables [9], although the output then forgets the early layers [10, 11].

This difference matters in practice, because hardware noise is often made unital on purpose. Randomized compiling [12, 13] averages the noise over random Pauli gates, which turns AD into its Pauli twirl. The twirl keeps the shrinking of AD but removes its pull toward $|0\rangle$, leaving generalized amplitude damping at infinite temperature. Comparing AD with its Pauli twirl therefore isolates the pull toward $|0\rangle$, the zero-temperature bias of the damping. A recent study [14] made this comparison and found that twirling degrades eigensolver solutions and reduces the range of outputs a model can fit, and that reversed damping performs worse than ordinary damping. What these effects mean is less clear. They could reflect a real loss in what the model can learn, or only a change in how its outputs must be read out. It is also unknown whether the direction of the damping matters in itself.

In this work, we show that in our models the zero-temperature bias acts mainly through the spread of the measured expectation values over the inputs. Under AD, this spread settles on a floor, whereas under the twirls it keeps shrinking with every layer. The floor vanishes as the temperature rises. A trainable readout absorbs most of the resulting differences in accuracy, much as rescaling methods of error mitigation do [15], so the twirled models lose little in what they can learn. The difference reappears instead as a cost in measurement shots [16]: at a fixed shot budget, the classifier under AD stays close to its noiseless accuracy, while the twirled classifiers fall far below it. At weak damping, the two separate only in very deep circuits.

We also show that the direction of the damping is often a gauge freedom. Reversing the damping changes the gates, the input state, and the measured observables as well. When the damping follows complete entangling layers and the circuit has trainable boundaries, these changes are absorbed by a Pauli frame or by spin time reversal, and reversing the damping amounts to a change of parameters. This extends the parameter symmetries of Fontana *et al.* [17] from one circuit to a pair of noise models. Time reversal leaves every single-qubit rotation unchanged, so only multi-qubit gates, the placement of the noise, and the circuit boundaries can make the direction physical.

We organize the text as follows. Section II introduces the channels and models. Section III treats the output scale, Sec. IV the gauge, and Sec. V the spread, the sampling cost, and their dependence on the damping strength. Section VI concludes.

II. CHANNELS AND MODELS

In this section, we introduce the noise channels that we compare and the three models on which we compare them: single-qubit re-uploading fits, a four-qubit classifier, and a three-qubit eigensolver.

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A. Channels

We describe each single-qubit channel $\mathcal{N}$ through its action on the Bloch vector, which is an affine map $\mathbf{r} \mapsto T\mathbf{r} + \mathbf{t}$ [18], with

$$T_{ij} = \tfrac{1}{2}\mathrm{Tr}\big[P_i\,\mathcal{N}(P_j)\big], \qquad t_i = \tfrac{1}{2}\mathrm{Tr}\big[P_i\,\mathcal{N}(I)\big], \tag{1}$$

where $P_i \in \{X, Y, Z\}$. The matrix T describes how the channel shrinks the Bloch vector, and the vector $\mathbf{t}$ how it pushes it. The channel is unital if and only if $\mathbf{t} = \mathbf{0}$.

We compare amplitude damping with two twirled versions of it. Amplitude damping of strength p does two things to the Bloch vector. It shrinks the vector, by $c = \sqrt{1-p}$ in the transverse directions and by c^2 along z, and it pushes the vector toward $|0\rangle$ by p. The Clifford twirl [19] averages these effects over all directions and leaves depolarizing noise, which shrinks every component by the same factor and pushes nowhere. The Pauli twirl [12, 20] is gentler. It keeps the shrinking of AD exactly but removes the push toward $|0\rangle$, the non-unital term.

The Pauli twirl can also be seen as an average over the two directions of the damping. Reversed damping, $\mathrm{AD}^{\mathrm{flip}}$, is AD conjugated by a bit flip and damps toward $|1\rangle$, and the twirl is

$$\mathrm{AD}_{\mathrm{Pauli}} = \tfrac{1}{2}\big(\mathrm{AD} + \mathrm{AD}^{\mathrm{flip}}\big). \tag{2}$$

This average has a simple physical meaning. Generalized amplitude damping [18] with excited-state population N has the same T as AD and $t_z = p(1-2N)$. AD, its Pauli twirl, and $\mathrm{AD}^{\mathrm{flip}}$ are the cases $N = 0$, $\frac{1}{2}$, and 1, so comparing AD with its Pauli twirl compares damping at zero and at infinite temperature.

We treat the twirled channels as idealized, replacing the channel by its twirl at every insertion. No twirling protocol can do this exactly after a non-Clifford gate such as the data encoding, so the twirled channels should be read as model channels that share the contraction of AD.

B. Single-qubit re-uploading fits

Our first model is the single-qubit data re-uploading model of Ref. [14]. With $\mathcal{U}(V)\colon \rho \mapsto V\rho V^\dagger$, the encoding $\mathcal{S}_x = \mathcal{U}(R_X(x))$, and trainable blocks $\mathcal{W}_\ell = \mathcal{U}(R_Z R_Y R_Z)$ with three parameters each, the noisy circuit with $L = 2$ layers is

$$\Phi_{x,\theta} = \mathcal{N}\circ\mathcal{W}_2\circ\mathcal{N}\circ\mathcal{S}_x\circ\mathcal{N}\circ\mathcal{W}_1\circ\mathcal{N}\circ\mathcal{S}_x\circ\mathcal{N}\circ\mathcal{W}_0, \tag{3}$$

and the raw model outputs

$$f(x) = \mathrm{Tr}\big[Z\,\Phi_{x,\theta}(|0\rangle\langle 0|)\big]. \tag{4}$$

To measure expressivity, we fit random targets and record the largest range R for which Adam reaches a mean squared error below 10^{-5} within 150 steps. As targets, we use 200 random degree-2 trigonometric polynomials, each rescaled to range R and the same for every channel. We then compare the raw model with a rescaled model $f_{s,b}(x) = s\,f(x) + b$, in which s and b are trained together with the circuit. Appendix D gives the remaining details of the protocol.

C. Four-qubit re-uploading classifier

Our second model is a data re-uploading classifier [21, 22] on $n = 4$ qubits (Fig. 1), which classifies MNIST images [23] of the digits $\{1, 3, 5\}$. Each image is reduced to four principal components, which are mapped linearly to angles $a_j \in [0, 2\pi]$. As a second data set, we use three classes of Fashion-MNIST [24].

The circuit is built from three ingredients, namely trainable blocks, the encoding, and a ring of CNOTs,

$$\begin{gathered}\mathcal{R}^{(\ell)} = \bigotimes_{j=1}^{n} R_Z R_Y R_Z, \quad S(\mathbf{a}) = \bigotimes_{j=1}^{n} R_X(a_j), \\ E = \mathrm{CNOT}_{n,1}\cdots\mathrm{CNOT}_{2,3}\,\mathrm{CNOT}_{1,2},\end{gathered} \tag{5}$$

where $\mathrm{CNOT}_{j,k}$ has control j and target k. Each layer applies a trainable block and the ring, then the encoding, with the channel acting after both,

$$\mathcal{L}^{(\ell)}_{\mathbf{a}} = \mathcal{N}^{\otimes n}\circ\mathcal{U}\big(S(\mathbf{a})\big)\circ\mathcal{N}^{\otimes n}\circ\mathcal{U}\big(E\,\mathcal{R}^{(\ell)}\big), \tag{6}$$

and the full circuit is

$$\Phi_{\mathbf{a},\theta} = \mathcal{N}^{\otimes n}\circ\mathcal{U}\big(\mathcal{R}^{(L)}\big)\circ\mathcal{L}^{(L-1)}_{\mathbf{a}}\circ\cdots\circ\mathcal{L}^{(0)}_{\mathbf{a}}, \tag{7}$$

with $2L+1$ insertions of the channel per qubit and θ the $3n(L+1)$ trainable angles of the blocks. The readout acts on the single-qubit expectation values

$$z_j(\mathbf{a},\theta) = \mathrm{Tr}\big[Z_j\,\Phi_{\mathbf{a},\theta}\big(|0\rangle\langle 0|^{\otimes n}\big)\big], \tag{8}$$

and the scores of the three classes are

$$\ell(\mathbf{a}) = \begin{cases} W\mathbf{z} + \mathbf{b} & \text{(raw)}, \\ W\,\mathrm{diag}(\boldsymbol{\sigma})^{-1}(\mathbf{z} - \boldsymbol{\mu}) + \mathbf{b} & \text{(standardized)}, \end{cases} \tag{9}$$

where W is a trainable $3 \times n$ weight matrix and $\mathbf{b}$ a trainable bias. In the standardized readout, $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ collect the means μ_j and the population standard deviations σ_j of the z_j over the 1000 training images. We recompute them at every training step, with gradients flowing through them, and apply their values at the final parameters to the test images. The standardized readout adds no parameters to the raw one. It only fixes the scale at which W acts.

Unless stated otherwise, we use $p = 0.3$ and simulate the density matrix exactly, so the expectation values carry no sampling error. The exception is the finite-shot runs, in which each z_j carries the noise of N_s measurements, in training as well as at test time. For these runs, we define the operational shot cost of a trained model as the smallest N_s at which its mean test accuracy lies

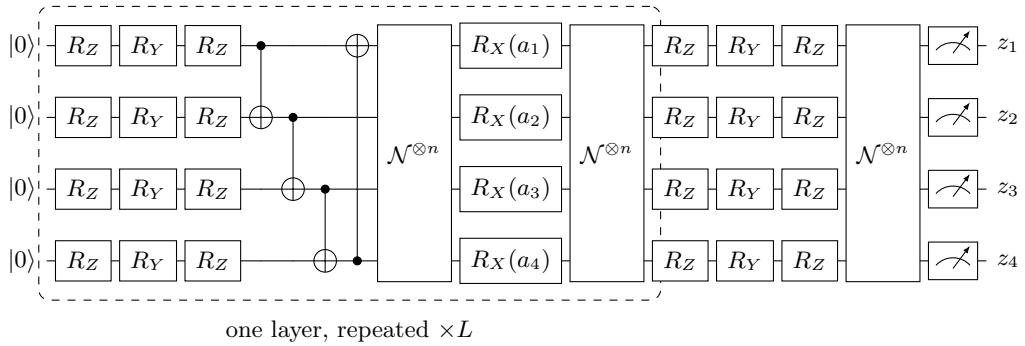


FIG. 1. The classifier at $n = 4$, drawn for one layer and the final block. Each $R_Z R_Y R_Z$ triple carries three trainable angles; the encoding column re-uploads the angle vector $\mathbf{a}$ in each of the L layers. The channel acts on every qubit after the entangler, after the encoding, and after the final block, giving $2L + 1$ insertions per qubit. The expectation values $z_j = \langle Z_j \rangle$ enter a trainable affine readout.

within 0.01 of its accuracy with exact expectation values, and we report its geometric mean over the initializations. Training settings, classical baselines, and the shot model are given in Appendix D.

D. Three-qubit variational eigensolver

Our third model is the eigensolver of Ref. [14] for the periodic transverse-field Ising chain on three qubits,

$$H = -J \sum_{i=1}^{3} Z_i Z_{i+1} - h \sum_{i=1}^{3} X_i, \tag{10}$$

with $J = 1$, $h = 0.5$, and $Z_4 \equiv Z_1$, whose ground-state energy is $E_0 = -3.2321$. The ansatz starts with a trainable R_Y on each qubit, followed by four blocks. Each block applies a trainable $R_{ZZ}(\phi)$ to each of the three pairs and a trainable R_X to each qubit, and each two-qubit gate is decomposed as

$$R_{ZZ}(\phi) = \mathrm{CNOT}_{i,i+1} \left[I \otimes R_Z(\phi) \right] \mathrm{CNOT}_{i,i+1}. \tag{11}$$

We treat single-qubit gates as noiseless and write the noise strength of the eigensolver as γ. Since the placement of the noise turns out to matter, we compare two placements with the same number of insertions. In their *inside* placement, the channel acts on both qubits after each CNOT. In the *after* placement, it acts twice on both qubits after the complete R_{ZZ}. For each noise strength and placement, we minimize the energy from 50 initializations, the same for every channel, and report the relative error of the energy of the output state $\rho(\theta)$ with respect to E_0. Optimizer settings and a cross-check against their code are given in Appendix D.

III. OUTPUT SCALE

We begin with the simplest way in which noise can change a variational model: by changing the scale of its output. AD and its Pauli twirl, damping at zero and at infinite temperature, share the same contraction matrix T. Any difference in scale between them must therefore come from the non-unital vector $\mathbf{t}$, which carries the zero-temperature bias (Sec. V). In this section, we ask how much of the effect of the noise a trainable output scale can undo, and what it costs to undo it.

A. Scale and sampling cost

Consider first the simplest case, in which the noise only rescales and shifts each expectation value,

$$\tilde{z}_j = s\, z_j + b_j, \tag{12}$$

with constants $s \neq 0$ and b_j that depend on neither the input nor the parameters. A trainable affine readout can then undo the noise completely: every classifier that it can build from the noiseless $\mathbf{z}$, it can also build from the noisy $\tilde{\mathbf{z}}$, so the noise leaves the model class unchanged. A single qubit under depolarizing noise is an exact example. Because this channel commutes with every unitary, N_{ins} insertions simply multiply the output by a constant,

$$\tilde{f} = \lambda_{\mathrm{depol}}^{N_{\mathrm{ins}}} f, \tag{13}$$

where $\lambda_{\mathrm{depol}} = \frac{1}{3}(2c + c^2)$ is the factor by which depolarizing noise shrinks each Bloch component, the average of those of AD. For anisotropic or non-unital channels, or on more than one qubit, the noisy model is no longer a rescaled copy of the noiseless one. Whether the scale still accounts for most of the difference then becomes an empirical question, which we test in the rest of this section. The average behavior of the spread on several qubits, its per-layer decay under the twirls and its floor under AD, is derived in Sec. V A and Appendix C.

On hardware, however, rescaling is not free. The number of shots is one of the main costs of variational algorithms [25, 26], and an expectation value estimated from

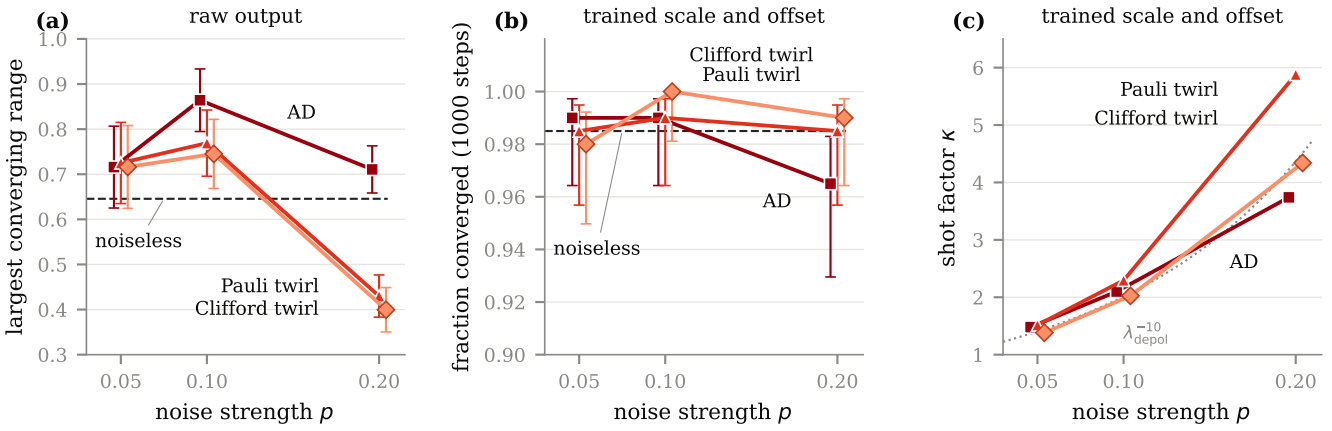


FIG. 2. Single-qubit re-uploading fits of Ref. [14] (200 targets). (a) Largest converging output range of the raw model $f = \langle Z \rangle$, with a free target offset. (b) Fraction of targets of range $R = 1$ that converge within 1000 steps when an output scale s and offset b are trained. (c) Shot factor $\kappa = (s/s_{\mathrm{None}})^2$ at $R = 1$ from the median fitted scale; the dotted line is the exact depolarizing value $\lambda_{\mathrm{depol}}^{-10}$. Bars are 95% confidence intervals of the mean over targets in (a) and Wilson 95% intervals in (b).

N_s shots, $\hat{z}_j$, has the variance

$$\mathrm{Var}(\hat{z}_j) = \frac{1 - z_j^2}{N_s} \leq \frac{1}{N_s}. \tag{14}$$

The standardized readout divides each z_j by its spread σ_j, so the shot noise of the standardized z_j has variance $(1 - z_j^2)/(N_s \sigma_j^2) \leq 1/(N_s \sigma_j^2)$. The smaller the spread, the more the shot noise is amplified. Keeping the class scores equally precise therefore requires $N_s \propto 1/\sigma^2$, and a channel $\mathcal{N}$ needs about $\kappa(\mathcal{N})$ times the shots of the noiseless circuit, with the shot factor

$$\kappa(\mathcal{N}) = \left(\frac{\sigma(\mathrm{None})}{\sigma(\mathcal{N})} \right)^2, \tag{15}$$

where σ is the spread σ_j at the trained parameters, averaged over the qubits and the initializations. For the single-qubit fits, we use $1/|s|$ in place of σ, with s the median fitted output scale, so that $\kappa = (s/s_{\mathrm{None}})^2$. In the language of error mitigation, the trained readout performs a rescaling [15, 27]. Such a rescaling cannot undo an exponential concentration without an exponential number of shots [16]. In Sec. V B, we test how well κ predicts the operational shot cost of Sec. II C, the number of shots that a trained classifier actually needs.

B. Single-qubit fits

We now examine the single-qubit fits. With the raw output and a free target offset, we recover the reported ordering at $p \geq 0.1$ [Fig. 2(a)]. At $p = 0.1$, for example, the largest converging range is 0.86 for AD, compared with 0.77 and 0.75 for the Pauli and Clifford twirls. This range is not a clean measure of expressivity, however. The noiseless model reaches only 0.65, below every noisy channel at $p \leq 0.1$. The metric therefore measures how quickly the model converges within the 150-step budget as much as the range it can reach. Indeed, when we center the targets on zero, the ordering at $p = 0.1$ reverses (Appendix B).

Once the model can train its own scale and offset, the difference between the channels disappears [Fig. 2(b)]. We train at two fixed ranges, $R = 1$ and $R = 0.468$, the latter being the median largest range of the noiseless raw model. At $R = 1$, AD converges within 1000 steps on 96.5 to 99.0% of the targets for $p = 0.05$ to 0.2, compared with 98.5 to 99.0% for the Pauli twirl and 98.5% for the noiseless model. None of the paired differences is significant at either range. Within 150 steps, the only significant difference appears at $R = 0.468$, and it favors the Pauli twirl (Appendix B).

What the twirl changes instead is the scale [Fig. 2(c)]. At $p = 0.2$, the Pauli twirl needs 1.24 to 1.57 times the shots of AD, depending on the range. The Clifford twirl needs 0.92 to 1.16 times the shots of AD, and its shot factor follows the exact depolarizing value $\lambda_{\mathrm{depol}}^{-10}$, since by Eq. (13) the five insertions shrink the output by $\lambda_{\mathrm{depol}}^{5}$. On a single qubit, the cost of twirling is therefore modest. As we show next, it becomes large in the classifier, where it compounds over qubits and layers.

C. Classifier

We now turn to the four-qubit classifier, where the effect of the scale is much stronger. With the raw readout and 100 training steps, the unital channels at first seem to lose trainability as the circuit gets deeper [Fig. 3(a) and Table III in Appendix B]. Between $L = 1$ and $L = 4$, the test accuracy falls from 0.847 to 0.715 for the Pauli twirl and from 0.851 to 0.695 for depolarizing noise, and the

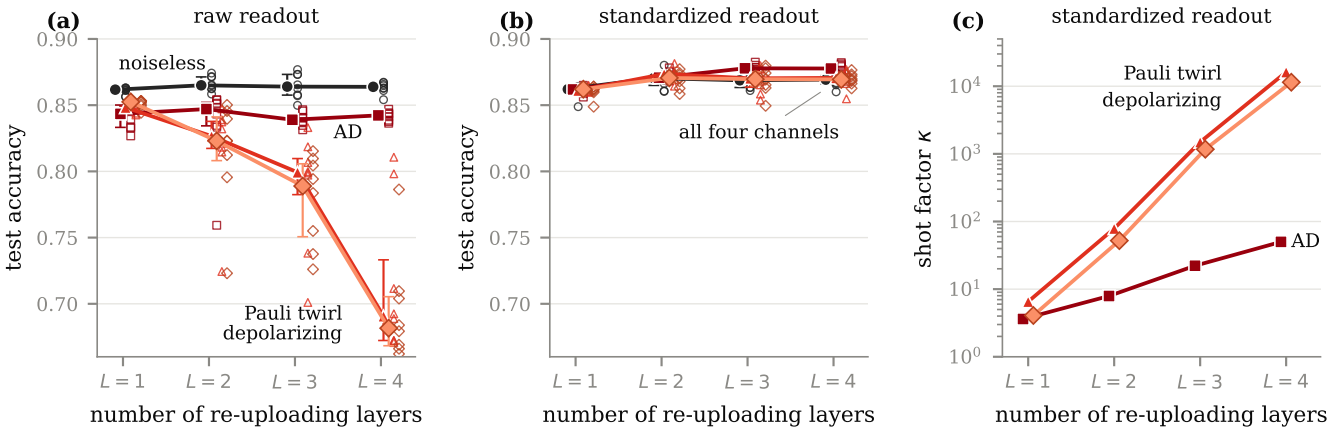


FIG. 3. Four-qubit classifier on MNIST against the number of re-uploading layers. (a) Test accuracy with the raw readout and (b) with the standardized readout, both after 100 steps, and (c) shot factor κ of Eq. (15) with the standardized readout. Panels (a) and (b) show the median and interquartile range over eight initializations, with hollow markers for the individual runs; one Pauli run at $L = 1$ in (b) diverged and lies below the axis.

training accuracy falls with it, to 0.700 and 0.672. AD stays at 0.84, and the noiseless circuit at 0.86. Taken at face value, this would suggest that the non-unital term keeps the deep circuit trainable.

A closer look shows that the drop comes from the readout, not from the circuit. If we train the same raw readout for 1000 steps, the unital channels recover to 0.837 and 0.826 at $L = 4$. Over these extra steps, the norm of their readout weights grows from about 19 to 106 and 109, while for the noiseless circuit it grows only from 8 to 20. The reason is that, after 100 steps, the unital channels have shrunk the spread to about 5 and 4 percent of the noiseless spread, and to below 1 percent before training. With a spread this small, the raw readout cannot grow its weights fast enough. The circuit itself remains trainable. With the standardized readout, every channel reaches the accuracy of the noiseless circuit at every depth [Fig. 3(b)], and the Pauli and depolarizing channels are statistically indistinguishable from it (Table II).

We see the same behavior on Fashion-MNIST at $L = 4$. With the raw readout, the unital channels fall 16 and 17 points below the noiseless circuit (0.765), and AD falls 5 points below it. With the standardized readout, all three channels come within two points of the noiseless circuit [Fig. 8 in Appendix B], while their shot factors are 34 for AD and 9.9×10^3 and 8.2×10^3 for the Pauli and depolarizing channels.

The depolarizing channel at $L = 1$ gives an exact check of this picture. Each z_j then crosses all three noise insertions as a weight-one Pauli string, so the channel multiplies every z_j by the same factor λ_{depol}^3, whatever the parameters. Since the standardized readout removes a uniform scale, the depolarizing and noiseless classifiers should make identical predictions, and indeed they do, on every test image. AD is not a pure rescaling, yet with the standardized readout it too stays within one percentage point of the noiseless accuracy at every depth (Table II).

What the standardized readout cannot remove is the difference in spread [Fig. 3(c)]. At the end of training, the shot factor of the unital channels grows by a factor of about 14 per layer, from 4.1 at $L = 1$ to 1.1×10^4 at $L = 4$ for depolarizing noise and to 1.6×10^4 for the Pauli twirl. For AD, it grows by only about 2.4 per layer, from 3.6 to 50. Note that the shot factor is not a property of the channel alone. It also depends on what the model is trained for. With the raw readout, the optimizer moves the parameters toward less contracted expectation values, and after 1000 steps at $L = 4$ the shot factors are 11 for AD, 320 for the Pauli twirl, and 470 for depolarizing noise.

In summary, the zero-temperature bias acts in our models mainly through the spread of the expectation values, and the reported loss of expressivity is a loss of scale. A trained readout absorbs this loss, but at a cost in shots that grows with depth.

IV. GAUGE FREEDOM OF THE DAMPING DIRECTION

So far, we have compared AD with its twirls, which remove the non-unital term altogether. A simpler comparison only reverses the term, by damping toward $|1\rangle$ instead of $|0\rangle$. In this section, we ask when such a reversal changes the model at all, and we test the answer in the classifier and in the eigensolver.

A. When the damping direction is a gauge

Amplitude damping toward $|1\rangle$, $\mathrm{AD}^{\mathrm{flip}}$ of Sec. II A, differs from AD only in the sign of $\mathbf{t}$. As we show next, whether a circuit can tell the two apart depends on how a Pauli frame propagates through it.

To see how, we write the reversed circuit as a sequence of elements, $\Phi^{\mathrm{rev}} = E_m^{\mathrm{rev}} \circ \cdots \circ E_1^{\mathrm{rev}}$, where the elements E_k^{rev} are its gates, encodings, and noise layers. The idea is to insert the identity, in the form $g_k \circ g_k = \mathrm{id}$, at every cut $k = 0, \ldots, m$, that is, before the first element, between neighboring elements, and after the last one. Each g_k is a Pauli frame [28], either a Pauli conjugation or a Pauli conjugation combined with complex conjugation in the computational basis,

$$g_k(\rho) = \begin{cases} P_k\, \rho\, P_k & \text{(unitary type)}, \\ P_k\, \bar{\rho}\, P_k & \text{(antiunitary type)}, \end{cases} \tag{16}$$

where P_k is a Pauli string and all g_k are of the same type. Grouping each element with its two neighboring frames, we can rewrite the reversed circuit as

$$\Phi^{\mathrm{rev}} = g_m \circ \tilde{E}_m \circ \cdots \circ \tilde{E}_1 \circ g_0, \tag{17}$$

$$\tilde{E}_k = g_k \circ E_k^{\mathrm{rev}} \circ g_{k-1}. \tag{18}$$

If every $\tilde{E}_k$ turns out to be an element of the original model, possibly with different parameters, the reversed model is the original one in disguise. Whether this happens can be checked element by element. To state the conditions precisely, we write $y(x;\theta)$ for the outputs of the model for an input x and parameters θ, that is, the expectation values it measures, and $y^{\mathrm{rev}}(x;\theta)$ for those of the reversed model.

Proposition 1 (Frame gauge)**.** *Reversing the damping is a reparametrization of the model,*

$$y^{\mathrm{rev}}(x;\theta) = \eta\, y(x;\theta') \tag{19}$$

for all inputs x, with a sign $\eta = \pm 1$ for each output and θ' depending only on θ, whenever a frame assignment of one type exists such that

(i) at every noise location the frame acts as X or Y on each noisy qubit;

(ii) every fixed gate maps the incoming frame to a Pauli frame;

(iii) every data-encoding gate is left invariant;

(iv) every trainable element is mapped back into its own family;

(v) the input state is absorbed by the first element on each qubit, and the final frame maps each measured observable to η times itself, with $\eta = -1$ allowed only if the readout has a learnable sign.

The proof is given in Appendix A.

Corollary 1 (Deterministic frames are gauge)**.** *Under the conditions of Proposition 1, two statements follow:*

(a) a deterministic relabeling of the noise by a Pauli frame or by time reversal leaves the reachable model class unchanged;

(b) if, in addition, $\theta \mapsto \theta'$ preserves the distribution of the initial parameters and the optimizer is equivariant under it, the trained models under AD *and* $\mathrm{AD}^{\mathrm{flip}}$ *are identically distributed.*

The idea of propagating Pauli operators through a parametrized circuit is not new. Fontana *et al.* [17] used it, as the σ-pulse method, to find parameter symmetries of a single model. They showed that these symmetries survive unital Pauli noise but can be broken by non-unital noise such as amplitude damping, because a frame that crosses the channel as X or Y maps AD onto $\mathrm{AD}^{\mathrm{flip}}$. Proposition 1 turns this observation around. Instead of relating two parameter settings of one noise model, it uses the same propagation to relate two noise models. It also adds the antiunitary frames and states the conditions on the encoding, the input, and the readout under which the map is a reparametrization.

The most useful antiunitary frame is spin time reversal,

$$\Theta = \bigotimes_q (-iY_q)\, K, \tag{20}$$

with K the complex conjugation of Eq. (16). Time reversal flips the sign of every Pauli matrix and of the imaginary unit. As a result, it leaves every single-qubit rotation $R_{\mathbf{n}}(a) = e^{-ia\,\mathbf{n}\cdot\boldsymbol{\sigma}_q/2}$, with $\boldsymbol{\sigma}_q = (X_q, Y_q, Z_q)$ the Pauli matrices of qubit q, unchanged. The damping, on the other hand, is reversed, because the Kraus operators of AD are real:

$$\Theta R_{\mathbf{n}}(a)\, \Theta^{-1} = R_{\mathbf{n}}(a), \qquad \Theta\, \mathrm{AD}\, \Theta^{-1} = \mathrm{AD}^{\mathrm{flip}}, \tag{21}$$

for any axis $\mathbf{n}$. This has a strong consequence. Condition (iii) holds for every single-qubit encoding, whatever its axis, so only the multi-qubit fixed gates, the placement of the noise relative to them, and the two boundaries of the circuit can break the gauge.

The single-qubit fits of Sec. III B show what happens when one of the conditions fails. These fits use general single-qubit blocks, an R_X encoding, noise after every gate, and the output $f = \langle Z \rangle$ directly. They satisfy every condition except (v). The frame left after the last channel flips the sign of $\langle Z \rangle$, and nothing in the model can absorb it. Reversing the damping therefore flips the sign of the output, up to the parameter map $\theta \mapsto \theta'$. The two model classes are mirror images of each other, and they differ as sets, because the last channel confines the output under AD and under reversed AD to

$$f(x;\theta) \in [2p-1,\, 1], \qquad f^{\mathrm{rev}}(x;\theta) \in [-1,\, 1-2p], \tag{22}$$

and both upper bounds are attained. The difference is nevertheless easy to remove: a target ensemble that is

TABLE I. Frame gauge for reversed amplitude damping, from an exhaustive search over all Pauli frames of each type (4^n strings; $n \leq 3$), with $p = 0.3$. "Gauge" means a frame assignment satisfying (i)–(v) exists; the explicit identity $y^{\mathrm{rev}}(x;\theta) = \eta\, y(x;\theta')$ is then checked on random parameters and inputs, and the largest deviation is quoted. "None" means no frame of that type satisfies (i)–(v). The eigensolver rows place the channel once after each whole R_{ZZ} or after each CNOT; applying it twice after each R_{ZZ}, as in Fig. 4(a), leaves the frame assignment unchanged.

model	Pauli frame	time reversal / antiunitary
classifier, R_X encoding, noise after ring	gauge, 2.2×10^{-16}	gauge, 3.3×10^{-16}
classifier, R_Y encoding	gauge (Y frame), 3.3×10^{-16}	gauge, 2.2×10^{-16}
classifier, R_Z encoding	none	gauge, 2.2×10^{-16}
classifier, generic axis	none	gauge, 3.3×10^{-16}
classifier, noise inside the ring	none	none
classifier, encoding before first trainable layer	none	none
single-qubit fits, $f = \langle Z \rangle$	none	none
with a learnable sign	gauge, $\eta = -1$, 1.1×10^{-16}	gauge, $\eta = -1$, 2.8×10^{-16}
eigensolver, noise after each R_{ZZ}	gauge, 2.8×10^{-16}	gauge, 4.4×10^{-16}
eigensolver, noise after each CNOT	none	none

symmetric under $f \to -f$, or a trainable output scale, makes the two channels statistically indistinguishable.

Corollary 2 (The Pauli twirl is not frame-equivalent to AD)**.** *Pauli twirling replaces* AD *by* $\frac{1}{4}\sum_{P\in\{I,X,Y,Z\}} P\,\mathrm{AD}\,P$*, which by Eq.* (2) *is a convex mixture of frame-equivalent channels. It is not itself frame-equivalent to* AD*: every (anti)unitary Pauli frame preserves* $\|\mathbf{t}\|$*, which is* p *for* AD *and* 0 *for the twirl.*

The frame gauge therefore cannot remove the difference between AD and its twirl. That difference is set by the non-unital term, of length $\|\mathbf{t}\| = p$, and we quantify its effect in Sec. V.

B. Classifier

We now look at the classifier of Fig. 1 in more detail. Besides the damping itself, time reversal changes three parts of it: the input state, the readout, and the ring of CNOTs. Each CNOT of the ring is mapped to

$$\Theta\,\mathrm{CNOT}\,\Theta^{-1} = -\mathrm{CNOT}\,(Z \otimes X), \tag{23}$$

that is, to itself up to a local Pauli residue on its input side, with Z on the control and X on the target. Because the noise acts only after the complete ring, these residues can be moved in front of the ring, where the preceding trainable blocks absorb them. The first block also absorbs $\Theta\,|0\rangle$, and the readout absorbs the sign of the expectation values. All five conditions of Proposition 1 therefore hold for R_X, R_Y, and R_Z encodings and for a generic encoding axis alike, and reversing the damping is an exact reparametrization, as we confirm to machine precision (Table I). Interestingly, a unitary Pauli frame is enough only when it commutes with the encoding generator and still flips the damping, which singles out X for R_X and Y for R_Y. For R_Z or a generic axis, only the antiunitary frame works. If the noise acts after each CNOT of the ring instead, the X residues would have to cross a noise location, and the exhaustive search of Table I finds no frame of either type.

Both conditions of Corollary 1(b) also hold. For each frame of Table I, every Euler angle of the trainable layers maps as $\theta_k \mapsto \pm\theta_k + m_k\pi$ with $m_k \in \{0, 1\}$, and the readout maps as $W \mapsto -W$ and $\mathbf{b} \mapsto \mathbf{b}$. These maps leave the initialization unchanged, because the model depends on each angle only modulo 2π and the Gaussian initialization of W is symmetric. The standardized readout maps $\boldsymbol{\mu} \mapsto -\boldsymbol{\mu}$ and $\boldsymbol{\sigma} \mapsto \boldsymbol{\sigma}$, and Adam without weight decay is equivariant under sign flips and translations of the coordinates. Trained classifiers under AD and $\mathrm{AD}^{\mathrm{flip}}$ are therefore identically distributed over initializations, and the same holds with shot noise, whose distribution is symmetric under $z \mapsto -z$. Comparing the two damping directions in our classifier thus carries no information about the sign of the non-unital term.

The training results agree with this. With the raw readout at $L = 4$, $\mathrm{AD}^{\mathrm{flip}}$ and AD reach nearly the same accuracy, and 96.7% of their individual predictions agree. The expected difference in accuracy is exactly zero. Their individual predictions differ only because each pair of runs starts from the same θ under both channels, rather than from θ and θ'. This makes the comparison a useful null test, because we know the answer in advance. A two-level bootstrap interval, which resamples both the initializations and the test images, includes zero, as it should, whereas an interval that resamples only the test images excludes it (Appendix B). Resampling the test set alone would therefore understate the uncertainty.

To find out what happens when the gauge is broken, we place the noise after each CNOT of the ring, where no frame exists, and train both directions with the standardized readout at $L = 4$. Since this variant applies the channel more often, we compare its two directions only with each other. They still agree within the uncertainty

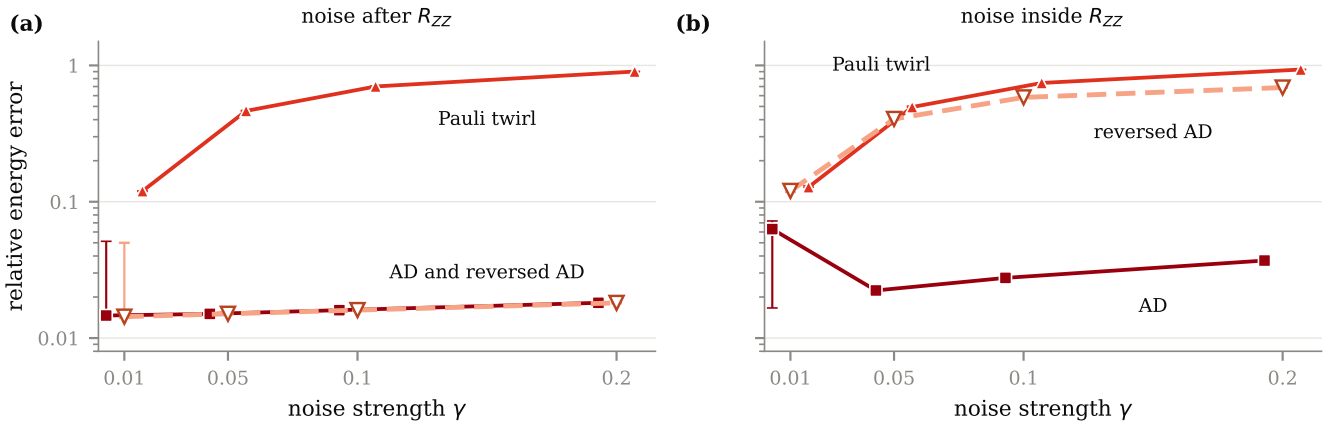


FIG. 4. Relative energy error $(E - E_0)/|E_0|$ of the three-qubit eigensolver against the noise strength, where $E = \mathrm{Tr}[H\rho(\theta)]$ is the energy of the optimized state, for AD, reversed AD (hollow markers), and the Pauli twirl (logarithmic vertical axis). Points are the median of 50 initializations and bars the interquartile range. (a) The channel acts after the complete R_{ZZ}; AD and reversed AD coincide. (b) The channel acts after each CNOT inside R_{ZZ}, as in Ref. [14]. At $\gamma = 0.01$, 20 (a) and 26 (b) of the 50 AD runs end in a higher local minimum, which widens the bars and in (b) raises the median above its value at $\gamma = 0.05$; from small initial angles, AD reaches 0.016 in (b).

in accuracy (Appendix B) and on 94.6% of the test images. Breaking the gauge thus allows a direction effect, but it does not guarantee a large one. In the classifier, the trainable layers and the readout may compensate for it. The eigensolver, with its fixed observable, cannot, as we show next.

C. Eigensolver

Finally, we turn to the eigensolver, where the direction effect was reported. Here the relevant symmetry is the global flip $X^{\otimes 3}$, which leaves the Hamiltonian H invariant. When the noise follows the complete R_{ZZ}, the pair flip $X^{\otimes 2}$ commutes with the gate. The flip then passes through every block, all the way back to the initial R_Y rotations, where it becomes a simple change of parameters,

$$E_{\mathrm{AD}^{\mathrm{flip}}}(\theta) = E_{\mathrm{AD}}(\theta'), \qquad \theta'_q = \pi - \theta_q, \tag{24}$$

where θ_q is the initial R_Y angle on qubit q and all other angles are unchanged. As a result, the optimized energies under AD and reversed AD coincide [Fig. 4(a)].

The situation changes when the noise acts after each CNOT. No frame can then flip the damping after both CNOTs of an R_{ZZ}. A frame that acts as X or Y on both qubits after the first CNOT acts as I or Z on the target qubit after the second, so the map fails (Table I). The optimized energies then separate [Fig. 4(b)], and the gap grows with γ, reaching 0.65 in relative error at $\gamma = 0.2$. We checked that this gap is a property of the energy landscape and not a failure of the optimizer. It persists for small initial angles, for the best of the 50 initializations, and for a second optimizer on an independent simulator (Appendix D). This reproduces the reported direction effect and traces it to where the noise sits inside the decomposed gate. It is also the case that matters in practice: on hardware whose native two-qubit gate is the CNOT, the damping acts at exactly this location.

Two further checks support this picture. On an open chain, flipping one sublattice maps J to $-J$ and is absorbed by the final R_X layer, so the sign of the coupling is itself a gauge. At $h = 2$ on the ring, the ground state is paramagnetic and far from every state that the final layer can prepare from $|000\rangle$. Even there, reversed AD still trails AD with the damping inside the gate, and AD still beats its twirl. Neither effect therefore rests on the ground state. Finally, the eigensolver shows the limit of rescaling. Unlike a classifier output, the energy cannot be rescaled freely, so here the twirl costs accuracy unless a mitigation step restores the scale [15].

V. WHAT THE NON-UNITAL TERM CHANGES

Sections III and IV showed that a trainable scale removes most of the difference between AD and its twirls, and that the direction of the damping is often a gauge. What remains, by Corollary 2, is the size of the non-unital term. In this section, we show what it changes: the spread of the expectation values before training, the number of shots needed after training, and how both depend on the damping strength.

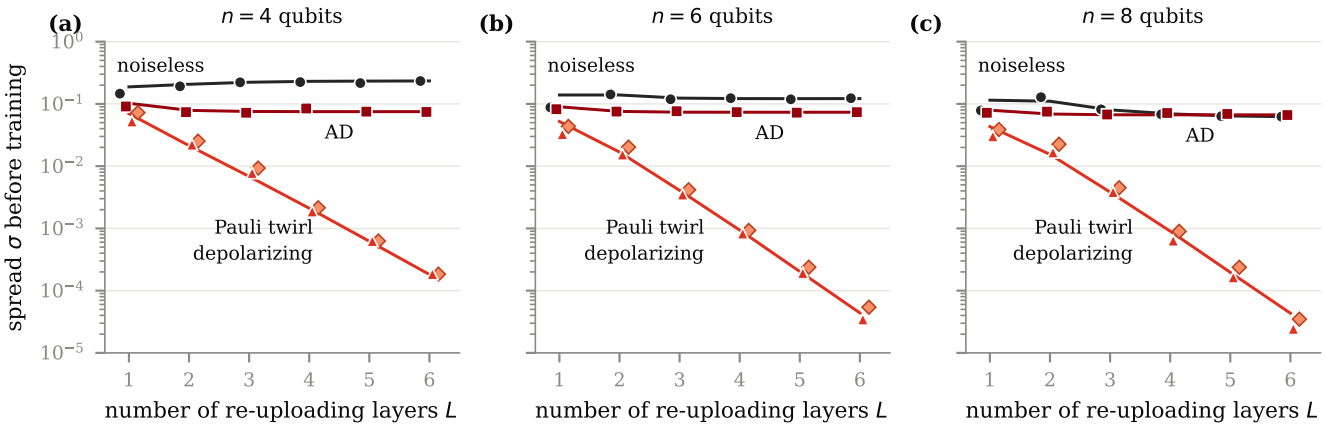


FIG. 5. Spread σ of the expectation values before training against the number of re-uploading layers at $p = 0.3$, for (a) four, (b) six, and (c) eight qubits, with n principal components as inputs. Markers are the root-mean-square spread over 300 training images and five parameter draws, shared across depths, for all four channels. Lines are the exact average over parameters for the noiseless circuit, AD, and the Pauli twirl (Appendix C).

A. Floor of the spread

We start before training, where the effect is cleanest, and ask why the spread under AD behaves differently from that under the twirls. Here we measure the spread as

$$\sigma^2 = \mathbb{E}_\theta \frac{1}{n} \sum_{j=1}^{n} \mathrm{Var}_{\mathbf{a}}\, z_j, \tag{25}$$

the variance of the z_j over the inputs, averaged over the qubits and over random parameters, and we write σ_{AD}, σ_{Pauli}, and σ_{None} for its values under AD, the Pauli twirl, and no noise. At $p = 0.3$, the spread separates the channels with both depth and width (Fig. 5). Under the unital channels, the spread shrinks by a nearly constant factor per layer at every width. Under AD, in contrast, it settles on a floor after two or three layers. As the circuit gets wider, the noiseless spread falls, as expected for a scrambling circuit, but the floor barely moves. At $n = 8$ and $L \geq 4$, the AD spread is therefore as large as the noiseless one.

The floor comes from the non-unital term. To see this, consider a Z component propagated backward from the readout. Under AD it becomes $c^2 Z + t_z I$, with $t_z = p$, whereas the Pauli twirl keeps only $c^2 Z$. The identity part behaves very differently from the rest. Its expectation value is 1 in every state, so it reaches the readout no matter what the earlier layers do. The twirl has no such part, and every component keeps shrinking from layer to layer. At strong damping, the branching into the identity after the last entangler sets the floor. Averaging its contribution over the parameters (Appendix C), we obtain the closed form

$$\sigma_\infty^2 = \frac{p^2(1-p)^2}{n} \sum_j \Big[\frac{(1-p)^2}{2} \mathrm{Var}(\cos a_j) + \frac{1-p}{4} \mathrm{Var}(\sin a_j) \Big], \tag{26}$$

which at $p = 0.3$ matches the floor of the AD spread to within 0.5% for $n \geq 4$.

This floor fits well with what is known about noise-induced concentration. Unital noise concentrates local expectation values exponentially in depth [7, 10], whereas under non-unital noise their variance over random circuits stays bounded from below by an amount set by the non-unital term [9]. Equation (26) is the counterpart of that bound for the variance over inputs, and both scale as $\|\mathbf{t}\|^2 = p^2$ without decaying with depth.

The floor also sets how the zero-temperature bias depends on temperature. At finite temperature, the damping of a qubit with splitting $\hbar\omega$ relaxes toward the thermal excited-state population $N = (e^{\hbar\omega/k_B T} + 1)^{-1}$, and at fixed p, that is, at fixed measured T_1, Sec. II A gives

$$t_z = p(1 - 2N) = p \tanh \frac{\hbar\omega}{2k_B T}. \tag{27}$$

The floor therefore falls linearly in $1 - 2N$ as the temperature rises and vanishes at $N = \frac{1}{2}$, the Pauli twirl (Appendix C). In summary, the identity part of the non-unital term keeps the AD spread on a floor at every depth and width, whereas the twirled spreads keep shrinking.

B. Sampling cost

We next ask whether the floor matters for the number of shots that a trained classifier needs, the operational shot cost of Sec. II C. To answer this, we train

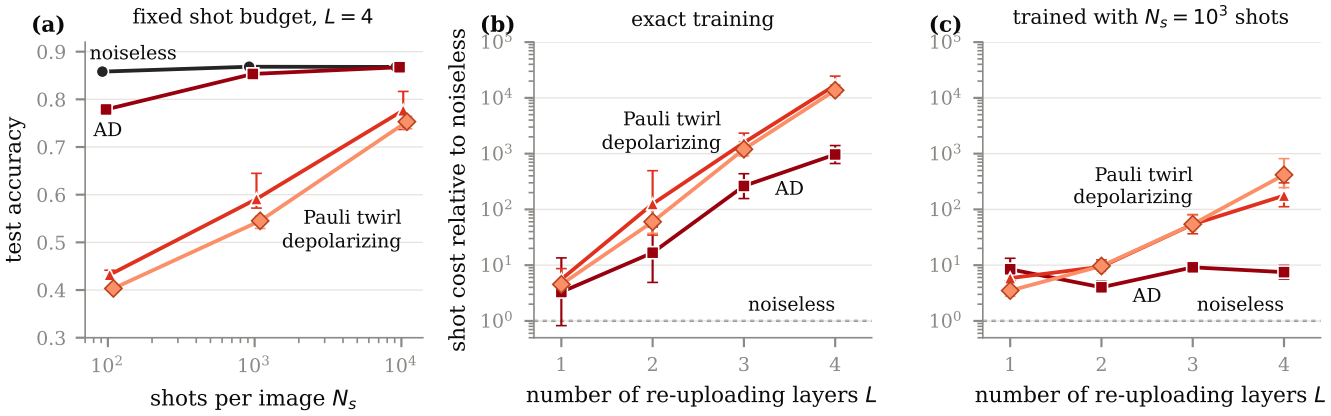


FIG. 6. Finite shots in the MNIST classifier at $p = 0.3$. (a) Test accuracy at $L = 4$ of models trained and tested with N_s shots per image (median and interquartile range over eight initializations). (b), (c) Operational shot cost relative to the noiseless circuit (geometric mean over initializations, with 95% bootstrap intervals over initializations) for models trained with exact expectation values (b) and with $N_s = 10^3$ shots (c).

under shot noise [Fig. 6(a) and (c)]. Trained and tested with 10^3 shots per image, AD stays within 1.5 percentage points of the noiseless accuracy at every depth, while the twirled channels lose up to 33 points at $L = 4$. Part of this loss already arises in training. Evaluated without shot noise, the twirled models trained with shots remain clearly less accurate than those trained exactly, whereas AD loses almost nothing. After training with shots, the operational cost of AD stays close to its shot factor κ, whereas for the twirls it still grows by a factor of 3 to 5 per layer [Fig. 6(c)].

The reason is that training under shot noise moves the circuit toward expectation values whose spread exceeds the shot noise $1/\sqrt{N_s}$. Under AD, the smallest spread rises to the floor of Sec. V A, above the shot noise, while under the twirls it stays below it (Appendix B).

Without shot noise in training, κ follows the cost of the unital channels but misses that of AD [Fig. 6(b)]. At $L = 4$, AD then needs 19 times more shots than its shot factor predicts, because the exactly trained readout relies on expectation values of very small spread, as a readout-weighted noise gain shows (Table IV in Appendix B). Even so, the Pauli twirl still needs 18 times the shots of AD. The floor thus matters for the shot budget. It lets a model trained under shot noise keep a cost close to κ, while the twirled models pay a cost that grows with every layer.

C. Dependence on the damping strength

So far we have worked at strong damping, $p = 0.3$. We now turn to weaker damping, closer to current hardware, and ask at which depth the difference between AD and its twirls appears. Both sides of the comparison change with p. The twirled spread decays more slowly per layer, and the floor of the AD spread gains a contribution from

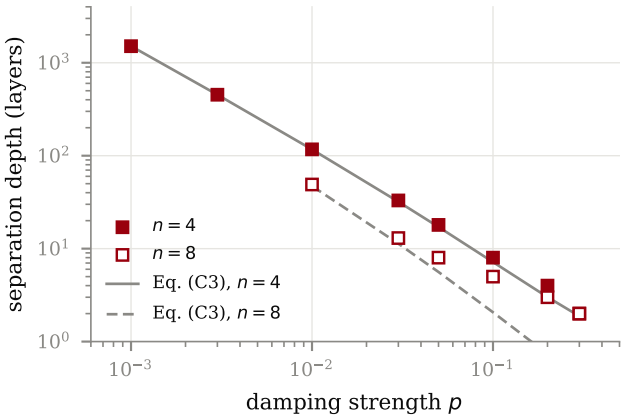


FIG. 7. Separation depth, at which the Pauli twirl needs ten times the shots of AD before training, against the damping strength p for four (filled) and eight (hollow) qubits. Markers are the exact average over parameters (5000 image pairs at $n = 4$, 700 at $n = 8$, computed for $p \geq 0.01$ at $n = 8$). Gray lines are the estimate of Eq. (C3) in Appendix C.

earlier layers (Appendix C). To quantify when the two spreads separate, we define the separation depth as the depth at which the squared ratio of the AD and Pauli-twirl spreads first reaches 10. There, the twirl needs ten times the shots of AD before training. At weak damping, the separation depth grows roughly as $p^{-1} \ln(1/p)$ (Fig. 7), in close agreement with a simple estimate that we derive in Appendix C.

Trained classifiers follow this picture (Table V in Appendix B). Well below the separation depth, AD and its twirls reach the same accuracy at nearly the same shot cost, and at the separation depth the twirls need about four times the shots of AD.

This has a clear consequence for real devices. On su-

perconducting hardware, the damping is of order 10^{-3} per two-qubit gate, for gate times of 30 to 400 ns and T_1 of order 100 μs [29]. In our model, one of the two insertions per layer follows a ring of n CNOTs, so p per insertion corresponds to several gate durations. At $p = 10^{-3}$, the separation needs about 1500 layers at $n = 4$ (Table VI in Appendix C). At realistic damping, the difference between AD and its twirls therefore appears only in very deep circuits.

VI. CONCLUSION

We have compared amplitude damping with its twirls, in particular with the Pauli twirl, which is damping at infinite temperature. In our models, the zero-temperature bias acts mainly through the spread of the expectation values. Under AD, the spread stops at a floor fixed at strong damping by the last layer, and at fixed T_1 this floor falls linearly in $1 - 2N$ as the temperature rises. Under the twirls, it keeps shrinking by a constant factor per layer at every width up to eight qubits. Training an output scale removes most of the resulting differences in fit and accuracy in exact simulation. What it removes reappears as a sampling cost: at a fixed shot budget, the classifier under AD stays close to its noiseless accuracy, while the twirled classifiers fall far behind. At weak damping, the two channels separate only after many layers, so the difference matters at strong damping or in very deep circuits.

The direction of the damping is a gauge, that is, reversing it changes only the parameters, whenever the damping follows complete entangling layers and trainable elements at both ends of the circuit absorb the flip. With a symmetric initialization and an equivariant optimizer, the trained models under the two directions are then identically distributed, as in our classifier. The direction can matter only where no frame can be absorbed, as in the eigensolver with the damping inside the decomposed two-qubit gate.

These results suggest three practices for comparing noise structures in variational models. First, report the output scale, or train one, together with the shot budget it implies. Second, before attributing an effect to the direction of the noise, check whether the model class is invariant under the relevant frames. Third, state where the noise acts relative to the decomposition of the gates.

Several questions remain open. Our simulations are exact and small: one qubit for the fits, four for the classifier, and three for the eigensolver. We established the width dependence before training, up to eight qubits, but training beyond four qubits was out of reach of exact simulation, and our shot model in training neglects the correlations between qubits. It also remains to be seen whether trained models follow the temperature dependence of the floor, and whether the results hold on hardware, where damping comes with dephasing.

TABLE II. Standardized-readout classifier on MNIST: difference from the noiseless circuit in test accuracy. P values from paired t tests over the eight initializations, Holm-adjusted over the 11 tests of the table.

L	channel	Δacc	95% CI	P
1	AD	+0.001	$[-0.003, +0.005]$	1
1	Pauli	−0.059	$[-0.178, +0.001]$	1
1	Depol	0	exact	–
2	AD	+0.003	$[-0.003, +0.009]$	1
2	Pauli	+0.004	$[-0.003, +0.010]$	1
2	Depol	+0.000	$[-0.004, +0.004]$	1
3	AD	+0.009	$[+0.003, +0.015]$	0.02
3	Pauli	+0.001	$[-0.007, +0.008]$	1
3	Depol	+0.000	$[-0.007, +0.007]$	1
4	AD	+0.008	$[+0.002, +0.013]$	0.004
4	Pauli	+0.001	$[-0.007, +0.009]$	1
4	Depol	+0.001	$[-0.006, +0.008]$	1

ACKNOWLEDGMENTS

We thank Vietnam National University Ho Chi Minh City (VNU-HCM) for its support. Claude Opus 5.5 (Anthropic) was used throughout the preparation of this work: to search and survey the literature, and to check its numerical claims and its characterizations of prior work against the underlying data and sources.

DATA AVAILABILITY

The code, the result files, and the scripts that reproduce every number and figure in this paper are available at https://github.com/Identity-Operator/zero-vs-infinite-T-damping and archived on Zenodo at doi:10.5281/zenodo.23053251 [30].

Appendix A: Proof of the frame gauge

a. Setting. A model on n qubits prepares $\rho_0 = |0\rangle\langle 0|^{\otimes n}$, applies elements $E_1, \ldots, E_m$, and reads out

$$y_i = \mathrm{Tr}\big[O_i\, \Phi(\rho_0)\big], \qquad \Phi = E_m \circ \cdots \circ E_1, \tag{A1}$$

possibly followed by a learned affine map. Each E_k is a fixed unitary $\mathcal{U}(U)$, a data encoding $\mathcal{U}\big(S(x)\big)$ with $S(x) = \bigotimes_q R_{\mathbf{n}_q}(x_q)$, a trainable unitary from a family $\mathcal{V}$ (general single-qubit blocks, or Pauli rotations $e^{-i\theta Q/2}$), or a layer of the single-qubit channel AD on a set S_k of qubits. The reversed model replaces every noise layer by $\mathrm{AD}^{\mathrm{flip}}$.

b. Frames. Let $\mathcal{P}(\rho) = P\rho P$ for a Pauli string P, and $\mathcal{K}(\rho) = \bar{\rho}$. On Hermitian operators $\mathcal{K}$ is real-linear, trace preserving and positive, $\mathcal{K}^2 = \mathrm{id}$, and for any channel $\mathcal{E}$ with Kraus operators $\{K_j\}$ one has $\mathcal{K} \circ \mathcal{E} \circ \mathcal{K} = \bar{\mathcal{E}}$,

the channel with Kraus operators $\{\bar{K}_j\}$. The frames of Eq. (16) are $g = \mathcal{P}$ for the unitary type and $g = \mathcal{P} \circ \mathcal{K}$ for the antiunitary type. In both cases $g \circ g = \mathrm{id}$, and

$$g \circ \mathcal{E} \circ g' \colon \ \{K_j\} \mapsto \{P\, K_j^{(t)}\, P'\}, \tag{A2}$$

with $K^{(t)} = K$ for the unitary type and $K^{(t)} = \bar{K}$ for the antiunitary type.

c. Telescoping. Eqs. (17) and (18) hold for any frames $g_0, \ldots, g_m$ of one type, since every inserted $g_k \circ g_k$ is the identity.

d. The conditions. (i) Consider a noise layer on S_k with $g_k = g_{k-1}$ acting as X or Y on each $q \in S_k$. The Kraus operators of AD are real, and $X\mathrm{AD}X = Y\mathrm{AD}Y = \mathrm{AD}^{\mathrm{flip}}$ (Sec. II A), so for both types

$$g_k \circ \mathrm{AD}^{\mathrm{flip}} \circ g_{k-1} = \mathrm{AD}. \tag{A3}$$

Off S_k the frame passes unchanged.

(ii) For a fixed unitary U, $g_k \circ \mathcal{U}(U) \circ g_{k-1} = \mathcal{U}\big(P_k U^{(t)} P_{k-1}\big)$, which equals $\mathcal{U}(U)$ if and only if

$$P_k \propto U\, P_{k-1}\, U^{(t)\dagger}. \tag{A4}$$

The frame thus propagates by conjugation and must remain a Pauli string. For Clifford U this holds for every P_{k-1} of the unitary type; for real Clifford U (CNOT) also of the antiunitary type.

(iii) For an encoding, $g_k = g_{k-1}$ and $P_q R^{(t)}_{\mathbf{n}_q}(x) P_q = R_{\mathbf{n}_q}(x)$ for all x. For the unitary type this requires $[P_q, \mathbf{n}_q \cdot \boldsymbol{\sigma}_q] = 0$. For the antiunitary type, $\overline{R_{\mathbf{n}_q}(x)} = \exp[+ix\,\overline{\mathbf{n}_q \cdot \boldsymbol{\sigma}_q}/2]$, so the condition is

$$P_q\, \overline{\mathbf{n}_q \cdot \boldsymbol{\sigma}_q}\, P_q = -\mathbf{n}_q \cdot \boldsymbol{\sigma}_q, \tag{A5}$$

which holds for $P_q = Y$ and every $\mathbf{n}_q$, since $\bar{X} = X$, $\bar{Y} = -Y$, $\bar{Z} = Z$, $YXY = -X$, $YZY = -Z$, and $YYY = Y$.

(iv) A general single-qubit block V is mapped to $P_{k,q} V^{(t)} P_{k-1,q} \in SU(2)$ for any pair of frames: such blocks absorb, and may change, the frame on their qubits. A Pauli rotation is mapped to

$$P\, e^{\mp i\theta Q^{(t)}/2}\, P = e^{-i\theta' Q/2}, \qquad \theta' = \pm\theta, \tag{A6}$$

if and only if $g_k = g_{k-1}$ and $PQ^{(t)}P = \pm Q$; it can carry a frame but not change it.

(v) By Eq. (17) the input is $g_0(\rho_0) = P_0 \rho_0 P_0$ (as $\bar{\rho}_0 = \rho_0$). If the first element on qubit q is a general block V, the pair $V^{(t)} P_{0,q}$ is itself a general block acting on $|0\rangle$; if it is R_X or R_Y, $X|0\rangle$ and $Y|0\rangle$ equal $R_{X,Y}(\pi)|0\rangle$ up to phase and are absorbed by a shift of the angle; otherwise $P_{0,q}$ must fix $|0\rangle$, i.e. $P_{0,q} \in \{I, Z\}$. At the output, using $\mathrm{Tr}[O\bar{\tau}] = \mathrm{Tr}[\bar{O}\tau]$ for Hermitian arguments,

$$\mathrm{Tr}\big[O\, g_m(\tau)\big] = \mathrm{Tr}\big[P_m O^{(t)} P_m\, \tau\big], \tag{A7}$$

for any state τ, which is $\eta\,\mathrm{Tr}[O\tau]$ if and only if $P_m O^{(t)} P_m = \eta O$.

e. Completing the proof. If (i)–(v) hold, every $\tilde{E}_k$ in Eq. (17) is an element of the original model with reparametrized trainable parameters, the input frame is absorbed by the first elements and the output frame contributes the sign η; hence

$$y_i^{\mathrm{rev}}(x;\theta) = \eta_i\, y_i(x;\theta') \tag{A8}$$

for all x, which is Eq. (19). □

f. Corollaries. Statement (a) of Corollary 1 follows because $\theta \mapsto \theta'$ is a bijection of the parameter space. For (b), write the map as $\theta' = D_\theta \theta + \pi\mathbf{m}$, with D_θ a diagonal sign matrix and $\mathbf{m}$ an integer vector, as for every frame of Table I, together with $W \mapsto W\,\mathrm{diag}(\boldsymbol{\eta})$ for the readout, where $\boldsymbol{\eta}$ collects the signs η_i. The training losses $\mathcal{C}$ under AD and $\mathcal{C}^{\mathrm{flip}}$ under $\mathrm{AD}^{\mathrm{flip}}$ then satisfy $\mathcal{C}^{\mathrm{flip}}(D_\theta\theta + \pi\mathbf{m},\, W\mathrm{diag}(\boldsymbol{\eta}),\, \mathbf{b}) = \mathcal{C}(\theta, W, \mathbf{b})$, so their gradients transform with D_θ and $\mathrm{diag}(\boldsymbol{\eta})$. An optimizer whose update commutes with these maps (gradient descent and Adam without weight decay exactly, SPSA in distribution) carries the trajectory from $(\theta_0, W_0, \mathbf{b}_0)$ under AD onto the trajectory from $(D_\theta\theta_0 + \pi\mathbf{m},\, W_0\mathrm{diag}(\boldsymbol{\eta}),\, \mathbf{b}_0)$ under $\mathrm{AD}^{\mathrm{flip}}$. If the map preserves the distribution of $(\theta_0, W_0, \mathbf{b}_0)$, the trained models are identically distributed.

For Corollary 2, write a qubit channel as in Eq. (1). Unitary Pauli conjugation and complex conjugation act on $(T, \mathbf{t})$ as

$$\mathcal{P} \colon \ (T, \mathbf{t}) \mapsto (DTD,\, D\mathbf{t}), \tag{A9}$$

$$\mathcal{K} \colon \ (T, \mathbf{t}) \mapsto (FTF,\, F\mathbf{t}), \tag{A10}$$

with D a diagonal sign matrix and $F = \mathrm{diag}(1, -1, 1)$; both preserve $\|\mathbf{t}\|$. The Pauli twirl has $\mathbf{t} = 0$, while $\|\mathbf{t}\| = p$ for amplitude damping.

g. Time reversal and CNOT. A Z residue of Eq. (23) passes through amplitude damping ($Z\mathrm{AD}Z = \mathrm{AD}$); an X residue does not. The residue must therefore reach a general trainable block without crossing a noise location, which is the case in our classifier (the trainable layer immediately precedes the ring) and fails when noise is inserted after each CNOT (Table I).

h. Scope. The proposition is a sufficient condition. The rows marked "none" in Table I establish that no frame of either type exists; they do not by themselves prove that the classes differ. For the single-qubit fits, the output ranges of Eq. (22) prove it. For the eigensolver at $\gamma = 0.3$, with L-BFGS-B from 12 random initializations and the channel applied once after each R_{ZZ} or after each CNOT, the lowest energies are equal where the gauge holds (-3.1770 for both) and differ by 1.98 where it does not (-3.087 against -1.109), which reproduces the direction of the reported effect. For the classifier with noise inside the ring, Sec. IV B measures the difference directly.

TABLE III. Test accuracy of the classifier on 3037 MNIST images (mean ± SD over eight initializations) against the number of re-uploading layers L. The raw readout is trained for 100 steps and, at $L = 4$, for 1000 steps; the standardized readout of Eq. (9) for 100 steps. †One of the eight runs diverged (test accuracy 0.382); the other seven lie between 0.860 and 0.865.

	raw readout					standardized readout			
	$L = 1$	$L = 2$	$L = 3$	$L = 4$	$L = 4$, 1000 steps	$L = 1$	$L = 2$	$L = 3$	$L = 4$
None	$.861 \pm .003$	$.866 \pm .005$	$.865 \pm .009$	$.863 \pm .004$	$.862 \pm .007$	$.860 \pm .005$	$.869 \pm .007$	$.868 \pm .006$	$.869 \pm .005$
AD	$.842 \pm .010$	$.835 \pm .032$	$.839 \pm .005$	$.841 \pm .004$	$.872 \pm .003$	$.861 \pm .003$	$.872 \pm .003$	$.877 \pm .004$	$.876 \pm .005$
Pauli	$.847 \pm .003$	$.817 \pm .039$	$.787 \pm .044$	$.715 \pm .057$	$.837 \pm .013$	$.802 \pm .170^{\dagger}$	$.873 \pm .005$	$.869 \pm .009$	$.870 \pm .007$
Depol	$.851 \pm .003$	$.814 \pm .041$	$.778 \pm .034$	$.695 \pm .041$	$.826 \pm .012$	$.860 \pm .005$	$.869 \pm .006$	$.869 \pm .010$	$.870 \pm .004$

Appendix B: Statistical tests

We collect here the hypothesis tests behind the comparisons in the main text. Classifier results are given as the mean and standard deviation (SD) over eight initializations with fixed data. Because all runs share one finite test set, a difference between two conditions is uncertain through both the initializations and the test images. Accuracy differences therefore come with a 95% confidence interval (CI) from a two-level bootstrap [31] (2×10^4 resamples) that resamples both, with the initializations resampled jointly for the two conditions since they share their initial parameters. They also come with the P value of the paired t test across the eight initializations. To avoid confusion with the noise strength p, we denote P values by a capital P. The P values of Table II are Holm-adjusted over its 11 tests. Two remain below 0.05, both for AD, at $L = 3$ and 4.

a. Classifier on Fashion-MNIST. These results are shown in Fig. 8. At $L = 4$, the differences from the noiseless circuit are $+0.017$, CI $[+0.007, +0.029]$, $P = 0.03$, for AD; $+0.005$, CI $[-0.004, +0.014]$, $P = 0.28$, for the Pauli twirl; and $+0.009$, CI $[-0.0001, +0.018]$, $P = 0.08$, for depolarizing noise (paired t, Holm-adjusted over the three channels).

b. Shot cost. At $L = 4$ with exact training, the operational shot cost of AD exceeds its shot factor 19-fold, with a 95% bootstrap interval over initializations of 12 to 30. The Pauli twirl needs 18 times the shots of AD on MNIST (10 to 29) and 36 times on Fashion-MNIST (17 to 85). Trained and tested with 10^3 shots, AD lies 0.7, 0.9, 1.2, and 1.5 points below the noiseless accuracy at $L = 1$ to 4, and at $L = 4$ it stays within 1.5 points also with 10^4 shots. Evaluated with exact expectation values, the twirled models trained with 10^3 shots are about 0.11 less accurate than those trained exactly, and AD about 0.01. At $L = 4$ and $N_s = 10^3$, the geometric mean of the smallest spread is 0.075 for AD, compared with 0.0065 under exact training and 0.24 for the noiseless circuit, and 0.005 and 0.003 for the Pauli and depolarizing channels.

The extra cost of AD after exact training comes from the readout. The shot factor only looks at the average spread of the z_j, whereas the readout amplifies the shot noise of each z_j by its own weight. Exact training lets the AD readout rely on expectation values whose spread is far smaller than in the noiseless circuit. We capture this effect with the readout-weighted noise gain

$$G = \Big\langle \sum_j \|\mathbf{w}_j\|^2 \, \frac{1 - z_j^2}{\sigma_j^2} \Big\rangle, \tag{B1}$$

where $\mathbf{w}_j$ is the column of W that multiplies z_j and the average runs over the test images. If the shot noise of different qubits is uncorrelated, G is N_s times the total shot-noise variance of the class scores. Table IV compares the exact-training costs with G and with the shot factor. For $L \geq 2$, G reproduces the measured cost ratios to within a factor of 1.6.

c. Weaker damping. Table V lists the trained classifiers at weaker damping, with the separation depth L^* of Appendix C. Well below it, at $p = 0.05$ and $L = 4$ ($L^* = 18$), AD and its twirls reach the same accuracy at nearly the same shot cost, as κ predicts. Closer to it, at $p = 0.1$ and $L = 4$ ($L^* = 8$), the twirls already need more shots than AD. At $L = L^* = 8$, the channels finally separate. The twirls then need about four times the shots of AD, in the same direction as the spread ratios before and after training. At this depth, AD is also more accurate than the Pauli twirl, by $+0.026$ in accuracy, CI $[+0.017, +0.035]$. It is not, however, more accurate than the best noiseless circuit (0.869 at $L = 2$ to 4). Evaluated with 10^3 shots per image, AD reaches 0.69 against 0.83 for the noiseless circuit.

d. Damping direction. With the noise after the whole ring and the raw readout, $\mathrm{AD}^{\mathrm{flip}}$ minus AD is -0.003 in accuracy, CI $[-0.008, +0.002]$, $P = 0.21$; an interval that resamples only the test images, $[-0.0056, -0.0009]$, excludes zero. With the noise after each CNOT of the ring (Sec. IV B), $\mathrm{AD}^{\mathrm{flip}}$ minus AD is -0.006 in accuracy, CI $[-0.013, +0.001]$, $P = 0.10$.

e. Single-qubit fits. Convergence of AD and of the Pauli twirl on the same 200 targets is compared with McNemar's exact test. Within 1000 steps, no difference is significant at either range ($P \geq 0.34$). Within 150 steps, the only significant difference is -13.5 percentage points at $p = 0.2$ and $R = 0.468$ ($P = 0.0013$), in favor of the Pauli twirl. All others have $P \geq 0.077$. For the raw model, the largest converging ranges of AD and the Pauli twirl are compared with the paired Wilcoxon test: with

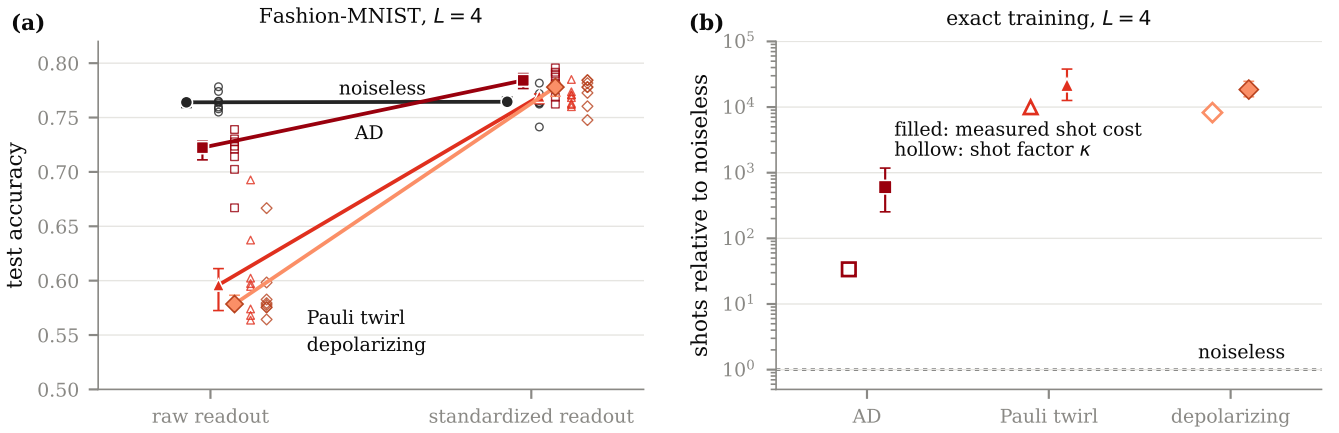


FIG. 8. Four-qubit classifier on Fashion-MNIST at $L = 4$ and $p = 0.3$. (a) Test accuracy with the raw and the standardized readout, as the median and interquartile range over eight initializations, with each run as a hollow marker. (b) Shot factor κ (hollow) and operational shot cost relative to the noiseless circuit (filled; geometric mean over initializations with 95% bootstrap intervals), both after exact training.

TABLE IV. Readout-weighted noise gain G of Eq. (B1) compared with the operational shot cost of the classifiers trained with exact expectation values at $p = 0.3$. All entries are relative to the noiseless circuit and are geometric means over initializations. For $L \geq 2$, G matches the shot cost to within a factor of 1.6, whereas the shot factor κ underestimates the cost of AD by up to 19 times.

		shot cost / noiseless			G/G_{None}			κ		
dataset	L	AD	Pauli	Depol	AD	Pauli	Depol	AD	Pauli	Depol
MNIST	1	3.3	5.6	4.5	0.9	1.3	4.3	3.6	6.5	4.1
MNIST	2	17	125	60	14	125	58	7.9	79	52
MNIST	3	263	1.6×10^3	1.2×10^3	277	2.0×10^3	1.6×10^3	22	1.5×10^3	1.2×10^3
MNIST	4	967	1.7×10^4	1.4×10^4	983	2.4×10^4	1.5×10^4	50	1.6×10^4	1.1×10^4
Fashion-MNIST	4	603	2.2×10^4	1.8×10^4	400	1.6×10^4	1.2×10^4	34	9.9×10^3	8.2×10^3

a free target offset, $P = 0.71$, 0.049, and 6×10^{-14} at $p = 0.05$, 0.1, and 0.2; with a centered target, $P = 0.19$, 9×10^{-7}, and 0.04.

f. Eigensolver. The relative errors of reversed AD and AD over the 50 initializations are compared with the Mann–Whitney test. With the noise inside R_{ZZ}, $P < 10^{-16}$ at every γ; with the noise after it, $P \geq 0.69$.

Appendix C: Spread before training

In this appendix, we derive the floor of the AD spread and its dependence on the damping strength, and we define the separation depth used in Sec. V C.

a. Exact average. The lines in Fig. 5 are exact averages, over parameters drawn uniformly from $[0, 2\pi)$, of Eq. (25), where the variance runs over the first 300 training images. We compute them in the Heisenberg picture from the second moments of the Pauli-string coefficients of Z_j, with the parameters shared between two images. Every uniformly random $R_Z R_Y R_Z$ block removes the cross moments between different strings because, averaged over its angles, the entries R_{bc} of its adjoint action have vanishing first moments and different columns are uncorrelated, $\mathbb{E}_\theta[R_{ab} R_{a'c}] = 0$ for $b \neq c$. Between two blocks the circuit acts site by site (channel, $R_X(a_j)$, channel), so the second moments of two images propagate through the elementwise product of the two composite site maps, followed by the entangling ring, which permutes the strings up to a sign. The second moments therefore obey a closed recursion, which is exact up to the sampling of image pairs: all pairs for $n \leq 5$, and 5000 random pairs for $n \geq 6$. Against brute-force averages of the density-matrix simulation, the recursion agrees within one to two standard errors in all 18 cells at $n = 2$ and 3 and $L = 1$ to 3. An independent density-matrix simulation at $n = 4$, with 200 parameter draws, agrees within two standard errors in all 12 cells for the noiseless circuit, AD, and the Pauli twirl at $L = 1$ to 4, and in all four cells for AD and the Pauli twirl at $p = 0.1$ and $L = 4$ and 8.

The measured points in Fig. 5 use five parameter draws, the same parameter vectors at every depth (those of depth L are the leading entries of those of depth 6), so

TABLE V. Trained classifiers at weaker damping (MNIST, standardized readout, exact expectation values, eight initializations): test accuracy (mean ± SD), operational shot cost relative to the noiseless circuit at the same depth, shot factor κ, and the squared ratio of the AD and Pauli-twirl spreads before training, $(\sigma_{\mathrm{AD}}/\sigma_{\mathrm{Pauli}})^2$, from the exact average over parameters of Table VI. The row $p = 0.3$ repeats Secs. III C and V B.

		test accuracy				shot cost / noiseless			κ			before
p	L	None	AD	Pauli	Depol	AD	Pauli	Depol	AD	Pauli	Depol	training
0.3	4	$.869 \pm .005$	$.876 \pm .005$	$.870 \pm .007$	$.870 \pm .004$	967	1.7×10^4	1.4×10^4	50	1.6×10^4	1.1×10^4	1.3×10^3
0.05	4	$.869 \pm .005$	$.866 \pm .005$	$.866 \pm .007$	$.864 \pm .004$	4.7	4.8	4.1	4.6	4.7	4.4	1.06
0.1	4	$.869 \pm .005$	$.872 \pm .005$	$.863 \pm .004$	$.863 \pm .003$	13.1	21.5	14.9	15.6	21.2	17.5	1.69
0.1	8	$.831 \pm .012$	$.866 \pm .006$	$.840 \pm .009$	$.838 \pm .007$	115	466	495	62	484	459	16.8

their deviations are correlated across L. The distribution over the parameters is heavy-tailed, and five draws tend to fall below the average, as for the Pauli twirl at $n = 8$ and $L \geq 4$. An independent Pauli-path simulation reproduces these five draws to 2×10^{-12} and, over 120 further draws at $L = 4$, agrees with the exact average within one standard error. In this exact average, the ratio of the AD and Pauli-twirl spreads grows by a factor of 3.3 per layer at $n = 4$ and 4.5 for $n \geq 6$, and its square, the ratio of their shot factors, by 11 and 20. At $n = 4$, the squared ratio is 2.1 at $L = 1$ and 1.3×10^3 at $L = 4$.

b. Floor at strong damping. Under AD the adjoint channel maps Z to $c^2 Z + pI$. The next random block averages the cross moments between the I and Z branches to zero. The AD spread therefore splits into the twirled part and a branching part,

$$\sigma_{\mathrm{AD}}^2 = \sigma_{\mathrm{Pauli}}^2 + \sigma_{\mathrm{br}}^2, \tag{C1}$$

where σ_{br} collects the paths through at least one branching into the identity. The damping after the last entangler branches the Z component of each z_j into the identity and adds

$$\delta z_j = p(1-p)\left[(1-p)\, v_z \cos a_j - c\, v_y \sin a_j\right] \tag{C2}$$

to z_j, where (v_x, v_y, v_z) is the direction into which the final trainable block rotates Z_j. Averaged over the final block, with $\mathbb{E}\, v_z^2 = \frac{1}{2}$, $\mathbb{E}\, v_y^2 = \frac{1}{4}$, and $\mathbb{E}\, v_z v_y = 0$, Eq. (C2) yields Eq. (26). The agreement within 0.5% quoted in Sec. V A uses all image pairs. For uniformly distributed angles, $\sigma_\infty = 0.096$. At strong damping, branchings in earlier layers are attenuated by the rest of the circuit, so the last layer alone sets the floor.

c. Weak damping. At weak damping, the twirled spread decays per layer as e^{-np}. This is because a scrambled Pauli string has about $3n/4$ non-identity sites, and each of the two insertions per layer shrinks such a site by a factor of about $1 - 2p/3$ on average. This decay rate agrees with np within 1% for $p \leq 0.05$ at $n = 4$ and 8. The floor of the AD spread at large depth, $\sigma_{\mathrm{br},\infty}$, changes as well (Table VI and Fig. 9). Its square has two parts, $\sigma_{\mathrm{br},\infty}^2 \approx \sigma_\infty^2 + b_{\mathrm{br}}\, p$. The first is proportional to p^2 [Eq. (26)]. The second comes from branchings in the roughly $1/(np)$ earlier layers that the damping does not yet suppress. It acts on the input dependence of those layers, which concentrates with width. For $p \leq 0.03$, b_{br} is $(1.8\text{–}2.7) \times 10^{-3}$ at $n = 4$ and $(1.0\text{–}2.4) \times 10^{-4}$ at $n = 8$, about an order of magnitude smaller.

d. Separation depth. We define the separation depth L^* as the depth at which $(\sigma_{\mathrm{AD}}/\sigma_{\mathrm{Pauli}})^2$ first reaches 10. There, the twirl needs ten times the shots of AD before training. By Eq. (C1), this happens when $\sigma_{\mathrm{br}} = 3\, \sigma_{\mathrm{Pauli}}$. With $\sigma_{\mathrm{br}} \approx \sigma_{\mathrm{br},\infty}$ and $\sigma_{\mathrm{Pauli}} \approx \sigma_{\mathrm{None}}\, e^{-npL}$, we obtain the estimate

$$L^* \approx \frac{1}{np} \ln \frac{3\, \sigma_{\mathrm{None}}}{\sigma_{\mathrm{br},\infty}}, \tag{C3}$$

where σ_{None} is taken at large depth (0.24 at $n = 4$ and 0.062 at $n = 8$). Equation (C3) agrees with the exact separation depth within 1% for $p \leq 0.01$ and within one layer for larger p at $n = 4$, and within two layers for $p \leq 0.03$ at $n = 8$ (Fig. 7). Since $\sigma_{\mathrm{br},\infty}$ falls roughly as a power of p and enters only through the logarithm, L^* grows roughly as $p^{-1} \ln(1/p)$.

e. Finite temperature. At finite temperature, the branching part of the spread at $n = 4$ follows $|1 - 2N|$ within 0.4% for every N and $L \leq 8$ at $p = 0.3$, and within 0.2% at $p = 0.1$. It is thus symmetric under $N \leftrightarrow 1 - N$, as the frame gauge of Sec. IV requires.

Appendix D: Training details and numerical checks

a. Single-qubit fits. Where Ref. [14] leaves a choice, we chose as follows. The 250 inputs lie on a uniform grid over $[-2\pi, 2\pi]$, and the target coefficients are drawn uniformly from $[-1, 1]$. Each target is shifted to zero mid-range and rescaled to range R on this grid. A step is one Adam update on a batch of 25 inputs, with a seeded shuffle renewed every ten steps and shared by all channels, and with their settings for $L = 2$ (learning rate 0.20, $\beta_1 = 0.45$, $\beta_2 = 0.95$). Initial parameters are drawn uniformly from $[0, 2\pi)$, the same for every channel. The largest converging range is found by bisection on $[0, 2]$ with ten halvings. Their free target offset is equivalent to a trainable b in $f = \langle Z \rangle + b$. The rescaled model starts from $s = 1$ and $b = 0$ and is trained for up to 1000 steps with the same Adam settings.

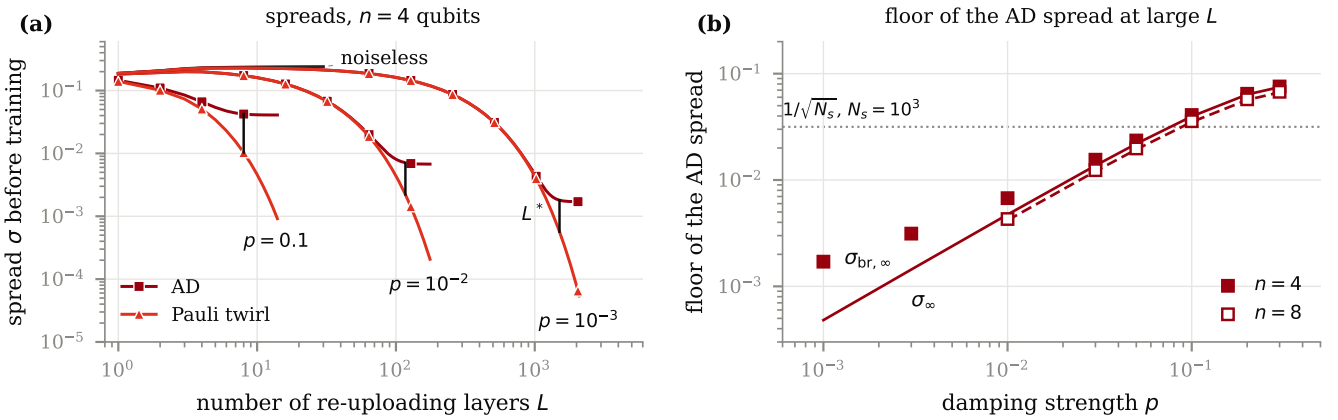


FIG. 9. Spread before training at weaker damping, from the exact average over parameters (5000 image pairs at $n = 4$, 700 at $n = 8$). (a) Spread against depth at $n = 4$ for AD (squares) and the Pauli twirl (triangles) at three values of p, and for the noiseless circuit up to $L = 30$; vertical segments mark L^*. (b) Floor $\sigma_{\mathrm{br},\infty}$ of the AD spread (markers) and its last-layer part σ_∞ [Eq. (26), lines] for $n = 4$ (filled, solid) and $n = 8$ (hollow, dashed); the dotted line is the shot noise $1/\sqrt{N_s}$ at $N_s = 10^3$. At $n = 8$ the exact average was computed for $p \geq 0.01$.

TABLE VI. Dependence on the damping strength p before training, from the exact average over parameters: the depth L^* at which $(\sigma_{\mathrm{AD}}/\sigma_{\mathrm{Pauli}})^2$ first reaches 10, the floor $\sigma_{\mathrm{br},\infty}$ of the AD spread and its last-layer part σ_∞ [Eq. (26)], and the number of shots $1/\sigma^2_{\mathrm{br},\infty}$ that resolves the floor. The floor is estimated from 5000 ($n = 4$) and 700 ($n = 8$) random image pairs, with a sampling error of about 1%.

	$n = 4$				$n = 8$
p	L^*	$\sigma_{\mathrm{br},\infty}$	σ_∞	$1/\sigma^2_{\mathrm{br},\infty}$	L^*
0.001	1508	0.0017	0.0005	3.4×10^5	–
0.003	453	0.0031	0.0014	1.0×10^5	–
0.01	117	0.0067	0.0047	2.2×10^4	49
0.03	33	0.0155	0.0137	4.2×10^3	13
0.05	18	0.0235	0.0219	1.8×10^3	8
0.1	8	0.0408	0.0396	600	5
0.2	4	0.0647	0.0637	240	3
0.3	2	0.0757	0.0747	170	2

b. Classifier training. To prepare the inputs, we standardize the pixel values, compute four principal components, and map each component linearly to an angle, sending its minimum to 0 and its maximum to 2π. The standardization, the principal components, and the extrema are fitted on 1000 training images and applied unchanged to all 3037 test images of the three digits. Training minimizes the softmax cross-entropy with Adam [32] at learning rate 0.05 for 100 full-batch steps unless stated otherwise, without weight decay. The circuit angles start uniformly in $[0, 2\pi)$, the readout weights from a zero-mean normal distribution with standard deviation 0.5, and the bias at zero. The density matrix is simulated exactly in PyTorch [33]. No regularizing constant is added to $\boldsymbol{\sigma}$ in the standardized readout. In one run, the spread of one z_j fell to 5×10^{-5}, and the training diverged (Table III). The Fashion-MNIST classes are T-shirt, pullover, and coat, with 3000 test images. For reference, logistic regression, a k-nearest-neighbor classifier ($k = 15$), and a support-vector machine with a radial kernel, trained on the same angles, reach 0.856, 0.879, and 0.886 on MNIST and 0.733, 0.758, and 0.748 on Fashion-MNIST. With the explicit parameter maps of Sec. IV B, the identity $y^{\mathrm{rev}}(x;\theta) = \eta\, y(x;\theta')$ holds to 9.4×10^{-16} for $n = 2$ to 4 and $L = 1$ to 4.

c. Shot model. For the finite-shot runs, each image is measured N_s times in the computational basis, and the estimate $\hat{z}_j$ of Eq. (14) is the mean of the outcomes ± 1 of qubit j. At test time we sample the outcomes from the exact distribution of the measured state, five times per model, and in every draw we re-estimate $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ from N_s shots of each training image, as one would on hardware. In training we replace each z_j by

$$\hat{z}_j = z_j + \sqrt{\frac{1 - z_j^2}{N_s}}\, \varepsilon_j, \tag{D1}$$

where ε_j is a standard normal random number, redrawn at every step and for every image, and compute $\boldsymbol{\mu}$ and $\boldsymbol{\sigma}$ from the noisy $\hat{z}_j$; gradients flow through z_j. This Gaussian model has the exact shot variance of each z_j but neglects the correlations between qubits and the shot noise of parameter-shift gradients. We train with $N_s = 10^3$ at $L = 1$ to 4, and with $N_s = 10^2$ and 10^4 at $L = 4$. The operational shot cost is interpolated in $\log N_s$ on a grid from 10 to 10^8. It is undefined for three exactly trained models at $L = 1$, which we omit: the diverged Pauli run, and one noiseless and one depolarizing model whose accuracy stays more than 0.01 below its exact value up to 10^8 shots.

d. Eigensolver. The eigensolver is trained with Adam at learning rate 0.05 for 500 steps, from parameters drawn uniformly from $[0, 2\pi)$. The reflection of Eq. (24) preserves the uniform initialization of the R_Y angles on $[0, \pi]$ used in Ref. [14], but their optimizer, COBYLA [34], builds its first simplex along the positive coordinate axes and is not equivariant under it, so for that optimizer only the reachable energies are guaranteed to coincide. For the inside placement, our implementation reproduces the energies of the open Qiskit [35] code [36] of Ref. [14] to 4×10^{-14} for the noiseless circuit, AD, reversed AD, and both twirls, at $\gamma \in \{0.01, 0.05, 0.1, 0.2\}$. With the damping after each complete R_{ZZ}, the two sides of Eq. (24) agree to 1.4×10^{-15} over 50 random parameter sets at every γ, and trajectories started from mapped initial parameters end at energies that agree to 6×10^{-15}. With the damping after each CNOT, the energies of the two channels at mapped parameters differ by 0.10 to 0.66 between $\gamma = 0.01$ and 0.2, and the median relative error of reversed AD exceeds that of AD by 0.06, 0.38, 0.56, and 0.65 at $\gamma = 0.01$, 0.05, 0.1, and 0.2. Small initial angles, with standard deviation 0.1 about zero, give the same gaps within 0.01 at $\gamma \geq 0.05$. The best of the 50 initializations for reversed AD lies within 0.01 of its median and above the median for AD at every γ. L-BFGS-B [37] from six further random initializations, run on an independently written simulator, reaches the same minima at $\gamma = 0.2$ (relative errors 0.0370 for AD and 0.6881 for reversed AD).

e. Coupling sign and field strength. On the open chain, with B the middle qubit, the energies of the two couplings agree at the mapped parameters to 2×10^{-15} for every channel and placement, and with $h = 0.5$ and $\gamma = 0.1$ reversed AD reaches a median relative error of 0.367 against 0.063 for AD for both $J = \pm 1$. At $h = 2$ on the ring and $\gamma = 0.1$, every state that the final R_X layer can prepare from $|000\rangle$ has a relative error of at least 0.54. There, AD reaches 0.29 and 0.23 with the damping inside and after the gate, against 0.59 inside the gate for reversed AD and 0.66 and 0.59 for the twirl. On the ring at $h = 0.5$ and $\gamma = 0.1$, the Pauli twirl gives the largest error in both placements, 0.74 inside and 0.70 after.

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